

# A Counterexample to Conjecture 00000002850: Nonnegative Rank Separates from Rank Already at Rank 3

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## Abstract

Conjecture 00000002850 asserts that the separation of nonnegative rank from ordinary rank “starts at rank 4”. It starts at rank 3: the  $4 \times 4$  slack matrix of a square has rank 3 and nonnegative rank 4. The lower bound on the nonnegative rank is a four-element fooling set. Everything is formalized in Lean 4 against Mathlib.

## 1 The conjecture as stated

The original text of conjecture 00000002850 reads:

Definition: Nonnegative matrix factorization (NMF): the algebraic geometry of non-negativity constraints. Conjecture: Rank rigidity of NMF: the separation of non-negative rank from ordinary rank starts at rank 4, with the minimal separating matrix an explicit  $4 \times 4$  nonnegative matrix; the geometric characterization of the separation is a non-triangulable pair of nested polytopes.

## 2 Reading of the statement

**Definition 1.** The *nonnegative rank*  $\text{rank}_+(M)$  of an entrywise nonnegative real  $m \times n$  matrix  $M$  is the least  $k$  such that  $M = WH$  with  $W$  ( $m \times k$ ) and  $H$  ( $k \times n$ ) entrywise nonnegative. Always  $\text{rank}_+(M) \geq \text{rank}(M)$ . We say the two ranks *separate at rank*  $r$  if some nonnegative matrix of rank  $r$  has nonnegative rank larger than  $r$ .

“The separation starts at rank 4” asserts that 4 is the least  $r$  at which the ranks separate: they separate at rank 4, and at no  $r < 4$ . The statement distinguishes this rank from the size of the minimal example (“an explicit  $4 \times 4$  matrix”), so the number 4 in “starts at rank 4” refers to the rank. We refute it by exhibiting a separation at rank 3.

## 3 Result

**Theorem 2.** *The slack matrix of a square*

$$S = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

has  $\text{rank}(S) = 3$  and  $\text{rank}_+(S) = 4$ . Hence the ranks separate at rank 3, and Conjecture 00000002850 is false.

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## 4 Proof

**The rank is 3.** The fourth row is  $r_1 - r_2 + r_3$ , so  $\text{rank } S \leq 3$ , and the leading  $3 \times 3$  block  $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$  has determinant 1, so  $\text{rank } S \geq 3$ .

**The nonnegative rank is 4.** Trivially  $\text{rank}_+ S \leq 4$ . Suppose  $S = WH$  with  $W, H \geq 0$  and inner dimension 3, so  $S_{ij} = \sum_{k=1}^3 W_{ik}H_{kj}$  is a sum of nonnegative terms. Each diagonal entry  $S_{ii} = 1$  is positive, so for each  $i$  there is a  $k = f(i)$  with  $W_{ik} > 0$  and  $H_{ki} > 0$ . If  $f(i) = f(j) = k$  for some  $i \neq j$ , then  $S_{ij} \geq W_{ik}H_{kj} > 0$  and  $S_{ji} \geq W_{jk}H_{ki} > 0$ . But for every pair  $i \neq j$  at least one of  $S_{ij}, S_{ji}$  is 0; the diagonal positions form a fooling set. So  $f: \{1, 2, 3, 4\} \rightarrow \{1, 2, 3\}$  is injective, which is impossible.

## 5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) states the conjecture and proves its negation against Mathlib.

- `HasNonnegFactorization M k` —  $M = WH$  with  $W, H$  entrywise nonnegative and inner dimension  $k$ .
- `SeparatesAt r` — some entrywise nonnegative real matrix  $M$  has `M.rank = r` (Mathlib’s rank) and no nonnegative factorization of inner dimension  $r$ .
- `ConjectureHolds` — `SeparatesAt 4` and  $\neg \text{SeparatesAt } r$  for all  $r < 4$ .
- `S, rank_S` — the matrix and its rank, via an explicit factorization through dimension 3 and the invertible leading block.
- `S_fooling, not_hasNonnegFactorization_S` — the fooling-set argument.
- `separatesAt_three, conjecture_00000002850_false`.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms:

```
#print axioms conjecture_00000002850_false
```

reports exactly `propext`, `Classical.choice` and `Quot.sound`.

## 6 Remarks

For matrices of rank at most 2 the two ranks coincide [1], so the separation starts exactly at rank 3, with  $S$  as a minimal example. Geometrically,  $S$  is the slack matrix of a square, and  $\text{rank}_+(S) = 4$  reflects the fact that a square in the plane cannot be written as the projection of a polytope with fewer than 4 facets, i.e. its extension complexity is 4 [2]. The “nested polytopes” picture of the statement belongs to rank 3, not 4: a rank-3 matrix has nonnegative rank 3 exactly when a triangle can be nested between the corresponding pair of polygons.

## References

- [1] J. E. Cohen and U. G. Rothblum, *Nonnegative ranks, decompositions, and factorizations of nonnegative matrices*, Linear Algebra Appl. 190 (1993), 149–168.
- [2] M. Yannakakis, *Expressing combinatorial optimization problems by linear programs*, J. Comput. System Sci. 43 (1991), 441–466.